

FinMMEval Lab
Multilingual and Multi modal Evaluation of Financial AI
Systems

Task 1: Financial Exam Q&A

George Mason University

Natural Language Processing (AIT-626-001)

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Introduction

Because there is so much financial information online now, like news articles, reports, and company announcements, it is getting harder to organize and understand financial text data by hand. It can be hard for computers to understand financial documents because they often have specialized terms, language specific to a certain field, and subtle meanings that depend on the context.

This project is based on Task 1 of the CLEF 2026 FinMMEval Lab. The goal of this task is to make models that can automatically sort financial text into the right domain categories. The job is to look at the content of financial texts and put them into one of several predefined financial domains, like finance, technology, taxes, or government regulation.

The main goal of this project is to classify financial domains, which means that the system needs to figure out what financial category a piece of text belongs in. This is an important job because putting financial data into structured categories makes it easier for analysts, investors, and regulatory bodies to find the information they need, make decisions, and keep track of what they know.

But this job is hard from a technical point of view for a number of reasons:

- Financial language often uses words and phrases that are only used in that field. But this job is hard from a technical point of view for a number of reasons:
- Financial language often uses words and phrases that are only used in that field.
- Text snippets can be short or unclear, which makes it hard to understand the context.
- Different fields may use the same words, like technology in fintech or government rules that affect financial markets.

To make an automated system that can sort financial text correctly, you need to know how to use natural language processing (NLP) and machine learning models that can figure out what words mean.

Related Work

The Natural Language Processing (NLP) community has done a lot of research on how to classify and analyze financial text. Usually, the job is to sort financial documents, news headlines, or reports into groups that have already been set up, like finance, technology, taxation, or regulatory topics. Naïve Bayes, Support Vector Machines (SVM), and Logistic Regression were some of the first methods used to represent text data. They used bag-of-words or TF-IDF features. These models did a good job of providing a baseline, but they often had trouble understanding context and using domain-specific language that is common in financial documents.

Recent improvements in deep learning and transformer-based language models have made financial NLP tasks much easier to do. BERT (Bidirectional Encoder Representations from Transformers) is one of the most important methods. It learns how to represent text in context by looking at words in relation to their context. Researchers have demonstrated that pre-trained language models can encapsulate intricate semantic relationships and enhance classification performance relative to conventional machine learning models.

Several models that are specific to certain fields have been made to help people understand financial text even better. FinBERT is a good example. It is a version of BERT that has been trained on a lot of financial documents, like analyst reports, earnings calls, and financial news. FinBERT has shown better results in tasks like financial sentiment analysis and text classification because it can pick up on specific financial terms and patterns that general language models might miss. Research indicates that FinBERT can surpass conventional machine learning techniques and even general-purpose BERT models in financial text analysis tasks.

The creation of financial benchmark datasets is another important step forward in financial NLP research. The MultiFin dataset, for instance, has multilingual financial news headlines with hierarchical labels that can be used for classification tasks. Researchers can use these kinds of datasets to test models in different financial areas and languages. This helps with research on both multi-class and multi-label classification problems.

In recent times, transformer-based architectures and large language models have emerged as the preeminent methodologies for financial text mining. Research comparing transformer models to conventional machine learning algorithms indicates that deep learning models more effectively capture intricate linguistic patterns and contextual meanings in financial texts. However, these models also have problems, such as high computational costs, difficulty in understanding the models, and the need for large datasets with labels.

The work in this project builds upon these existing research efforts by exploring financial domain classification using a publicly available financial dataset. A baseline model using traditional text representation and machine learning methods is implemented first, which provides a benchmark for evaluating more advanced models such as transformer-based architectures in later stages of the project.

Abstract

In this project, we mainly focus on multilingual financial domain for Task 1 of the CLEF 2025 FinMMEval lab where our goal is to automatically categorize short financial texts into domain classes such as Finance , Technology , Tax and accounting, business and management , goverment and controls and industry. In this earlier, we usually built a baseline using traditional NLP preprocessing and machine learning , After observing class imbalance and there are short text challenges in english dataset , we also worked on expanding the dataset by adding greek , spanish from hugging face.

In this checkpoint, we implemented an improved multilingual pipeline that mostly uses language native embeddings instead of translating everything to english. We also worked on generating sentence Bert multilingual embeddings for English, Greek and spanish texts. We also worked on fine tuning FinBERT per language to compare with other models. Our results show that SBERT embedding mostly improve performance for Logistic Regression and XG Boost across various languages. The final systems has data visualizations, Exploratory analysis, model training , confusion matrices and demo interface for predictions.

Keywords (5–8)

Multilingual NLP, Financial text classification, Sentence-BERT embeddings, Logistic Regression, XGBoost, FinBERT fine-tuning, Domain classification, CLEF FinMMEval

Objectives

The main goals of this project are:

1. Learn about the financial domain classification task that is part of the CLEF 2026 FinMMEval benchmark.
2. Look into and analyze the financial text dataset that was given to you. This includes finding important features and figuring out how the data is organized.
3. Do preprocessing and exploratory data analysis (EDA) on the text data to get it ready for modeling.
4. Create a basic machine learning model that can sort financial text into groups that have already been set.
5. Use the right evaluation metrics to see how well the baseline model works.
6. Find possible problems and areas for improvement that can help you make your models better in the future.

Selected Dataset

The dataset used in this project is derived from the FLARE-MultiFin English dataset, which is publicly available on Hugging Face:

Dataset source:

<https://huggingface.co/datasets/TheFinAI/flare-multifin-en>

...	id	query	answer	text	choices	gold
0	multifineng0	In this task, you're working with English head...	Finance	Deal summary Helen Oy / Infratek Finland Oy	[Finance, Technology, Tax & Accounting, Busine...	0
1	multifineng1	In this task, you're working with English head...	Finance	Deals	[Finance, Technology, Tax & Accounting, Busine...	0
2	multifineng2	In this task, you're working with English head...	Technology	PwC named a Major Player in the IDC MarketScap...	[Finance, Technology, Tax & Accounting, Busine...	1
3	multifineng3	In this task, you're working with English head...	Tax & Accounting	Proposal for digital services tax	[Finance, Technology, Tax & Accounting, Busine...	2
4	multifineng4	In this task, you're working with English head...	Business & Management	Social Impact Lab	[Finance, Technology, Tax & Accounting, Busine...	3
...	...	...	...	...	...	...
541	multifineng541	In this task, you're working with English head...	Business & Management	Generational transfer	[Finance, Technology, Tax & Accounting, Busine...	3
542	multifineng542	In this task, you're working with English head...	Tax & Accounting	Tax reporting & strategy	[Finance, Technology, Tax & Accounting, Busine...	2
543	multifineng543	In this task, you're working with English head...	Tax & Accounting	Taxes in Iceland	[Finance, Technology, Tax & Accounting, Busine...	2

Query Example - In this task, you're working with English headlines from the MULTIFIN dataset. This dataset is made up of real-world article headlines from a large accounting firm's websites. Your objective is to categorize each headline according to its primary topic. The potential categories are 'Finance', 'Technology', 'Tax & Accounting', 'Business & Management', 'Government & Controls', and 'Industry'. Your response should only include the category that best fits the headline.

Text: Deal summary Helen Oy / Infratek Finland Oy

The dataset consists of financial text samples collected from real-world financial sources such as news headlines, deal summaries, and financial reports. Each sample contains textual information along with a labeled financial domain category.

The dataset used in this project contains 546 samples and the following key fields:

Id, query, text, answer, choices and gold.

Domain Categories

Each text sample belongs to one of several financial categories, including:

- Finance
- Technology
- Tax & Accounting
- Business & Management
- Government & Controls
- Industry

These categories represent different areas of financial information and are used as target labels for classification. The dataset is suitable for supervised learning, where the goal is to train a model that learns patterns from labeled financial text and predicts the correct category for unseen examples.

The System Architecture / Framework

In this diagram, we have showed there is the end to end pipeline of our multilingual financial domain classification system. We started by collecting three datasets, FLARE mulfin parquet, Greek Plutus MultiFin and Spanish Flare ES MultiFin, then standardize each columns, clean the text and map greek spanish labels into common set of english domain labels.

We then worked and ran exploratory analysis to understand the class imbalance and text length pattern, For modeling, we compare three branches , A TF-IDF baseline with Logistic Regression, XGBoost, Our proposed multilingual embedding approach mainly uses SBERT embeddings with feature processing, followed by Logistic regression and XG boost and a transformer based approach where FinBert is fine tuned using a weighted loss trianer to handle imabalance. Finally we evaluate each model using accuracy, mcaro F1, confusion matrices and summarize all results.

SW HW Platforms:

Code Platform: Google Colab

Language: Python 3

Key Libraries/Frameworks:

Data + utilities: pandas, numpy, re, time

Visualization: matplotlib, seaborn

Datasets: datasets (Hugging Face)

Embeddings: sentence-transformers (SBERT multilingual)

ML models: scikit-learn (LogReg, scaling, SelectKBest, metrics), xgboost

Deep learning: torch

Transformers: transformers (FinBERT fine-tuning with Trainer)

Hardware: CPU/GPU via Colab



Baseline Solution:

Initially, we saw this as a text classification problem using traditional NLP features, the pipeline we used was simple and easy to reproduce. During the earlier stages, we worked on converting text to lowercase, removing punctuation, doing basic tokenization, created a clean text column and checking on class imbalance.

Baseline feature representation:

We basically used Term frequency Inverse Document Frequency for converting all these text into numerical vectors. The TF IDF mostly captures the presence of each word and it's importance by counts and weights.

The Baseline models we used are:

Logistic Regression:

- We used a Simple baseline classifier for multi class classification.
- We gave a decent accuracy but it did struggle with minority classes.

XGBoost:

- Gradient-boosted decision trees on TF-IDF vectors.
- There are Slightly improved handling of complex patterns compared to Logistic regression

FinBert:

- Used ProsusAI/finbert with Hugging Face Trainer, Fine-tuned on the English dataset with tokenization.
- Added ignore_mismatched_sizes=True because FinBERT originally has 3 labels.

Limitations:

- Mostly the texts are very short context and headlines. The dataset is very imbalanced , where minority class performance is weaker.
- The TF IDF is very much at surface level and doesn't work well with semantics.
- When we added more data, the performance dropped for combined dataset.

Proposed Solutions

To overcome the limitations observed in the baseline models, a series of improvements were implemented.

Multilingual Sentence-BERT Embeddings

This time, instead of translating everything into English, we used a multilingual embedding model that can work well with English, Greek and Spanish in same vector space.

Here the model we used is paraphrase-multilingual-mpnet-base-v2 which helps to capture semantic meaning not just word counts.

- We loaded the parquet file for english, added Greek HF dataset , Spanish Dataset.
- We worked on normalizing labels by mapping Greek , Spanish label names into categories using LABEL_MAP.

- Performed Cleaning , kept punctuation , sentence structure because SBERT benefits from that.
- Added Embeddings for each language to analyze each language's behaviour.
- Used PCA 768 - 2 dimension to visualize clusters by classes.

Feature Selection on Embedding for Logistic Regression and XG Boost.

After performing SBERT embeddings, we then performed the following,

- StandardScaler for normalizing features.
- Used SelectKBest to select top 300 embedding dimensions most correlating with label.
- For stratified train-test split we removed classes with less than 2 samples.

Training Models:

We then trained following models in each language,

- Logistic Regression
The class weight was balanced to reduce imbalance issues with parameters $C=2$, $\text{max_iter}=2000$.
- XGBoost:
We tuned hyperparameters such as $\text{n_estimators}=300$, $\text{max_depth}=6$, $\text{learning_rate}=0.1$, $\text{subsample}=0.8$, $\text{colsample_bytree}=0.8$.

This makes sure that these models no longer need to depend on raw word overlap. These models can learn from semantic embedding features.

FinBERT fine tuning

In this section, we fine tuned FinBert separately on English , Greek and Spanish dataset.

I used max length 256 and used a custom weighted trainer with class weighted cross entropy to handle these imbalance.

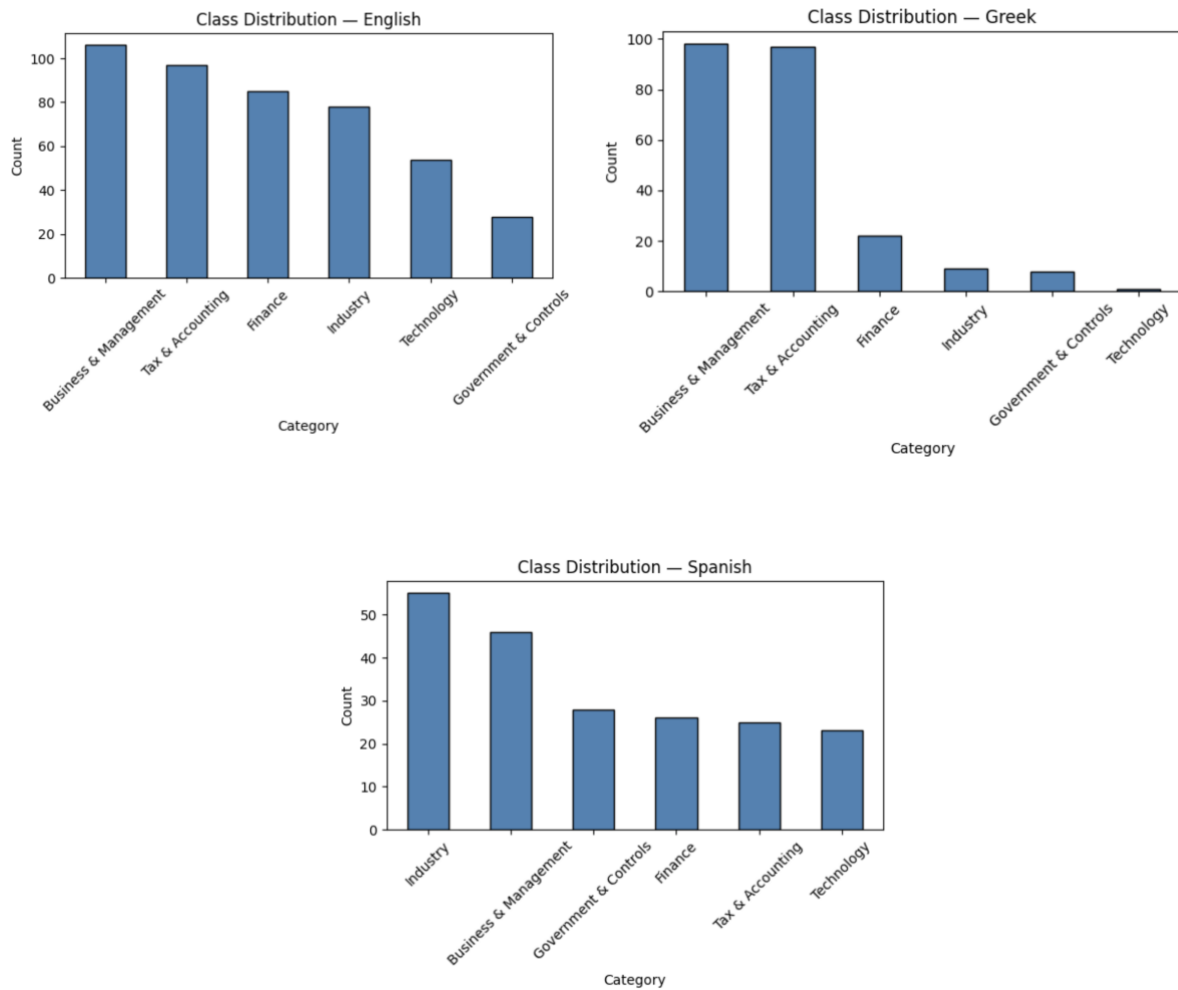
Drawbacks:

This FinBert underperformed on Greek or Spanish in our results as FinBert was originally planned on English Financial text, hence cross language adaptation is weak with a multilingual transformer.

SBERT embeddings handled Multilingual input as it is better due to embedding model multilingual by design.

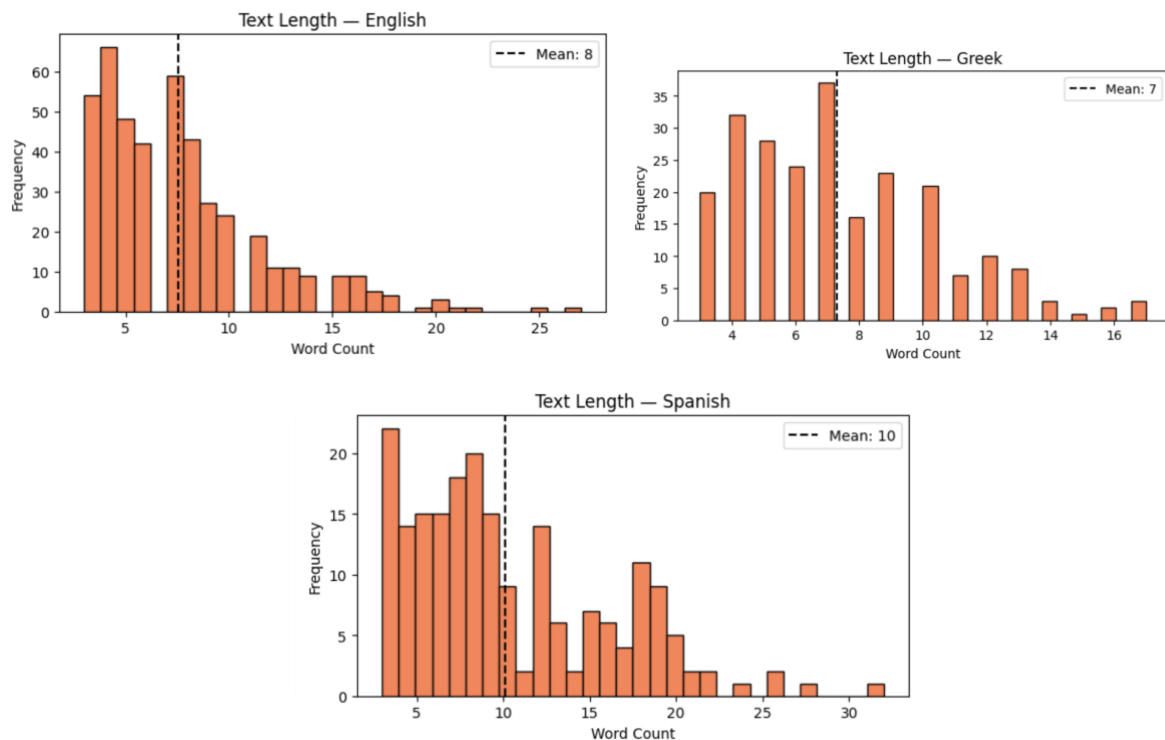
Exploratory Data Analysis and Interpretation

Class Distribution per Language:



In this figure it shows the number of samples in each domain categories for each and every language dataset. In English, Business and Management with 106 and Tax with 97 are highest. Government & Controls with 28 is the lowest which shows the dataset is imbalanced. In Greek, the imbalance is much more stronger such as Business Tax dominating where technology words has only one sample which makes the model look good on big classes. In Spanish the classes are evenly spread with Industry being highest and Technology being lowest.

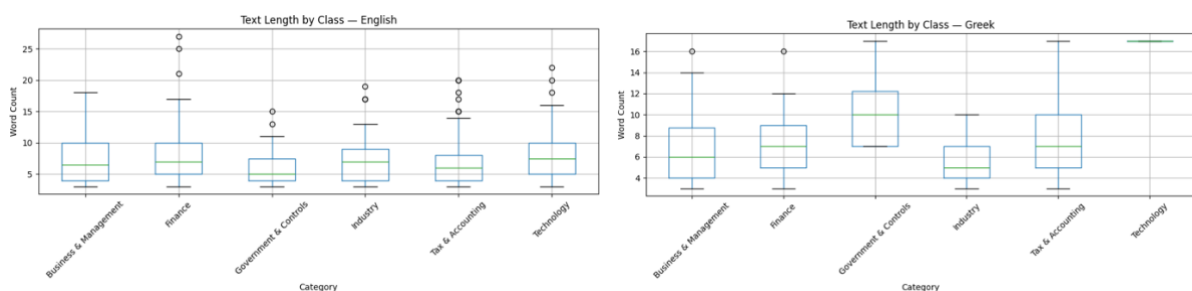
Text Length Distribution per Language

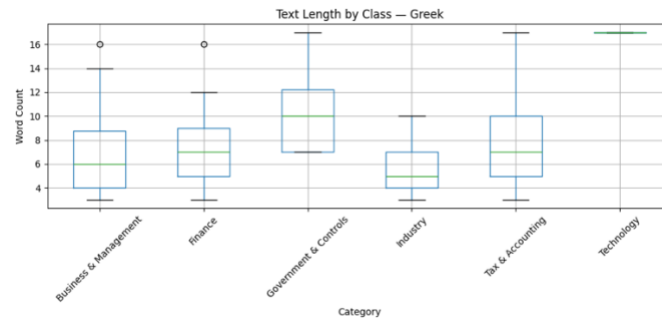


In these figures as mentioned above it shows about distribution of word counts for each language. All these three languages are mostly short texts but mean is different such as English has 8 words, Greek has 7 words, Spanish has 10 words. Spanish has longer entries and longer trail. Short texts usually give less context so models mostly get confused with categories and share similar keywords. In Text language English, the frequency is high for word count 5 and 10 which is same as Spanish. In Greek, for word counts between 4 and 8, the frequencies are high.

Text Length by Class per Language:

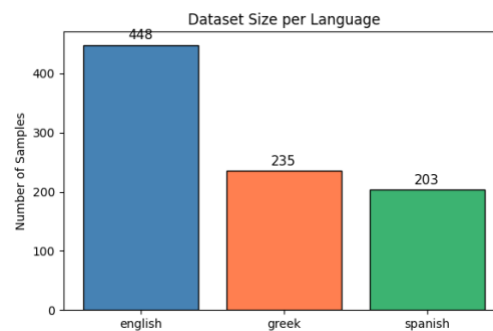
In the below graphs, these are boxplots which compare text length across various classes. In English, most class medians are so close but Finance and Technology show more outliers which means some samples are very much longer and may be easier to classify due to extra context. In case of Greek, the plot is less stable because it has more classes which have very few samples, so the distribution isn't much reliable. In Spanish the classes show much wider spreads and more longer outliers which matches the histogram as Spanish tends to have long text.





Dataset Size per Language:

The above graph shows how many cleaned samples we kept after filter all duplicates and removing , the english text has 448 , Greek has 235 , Spanish has 203. This shows why English has strongest training base which makes sure it leads to better patterns while Greek and Spanish results can fluctuate a lot due to smaller size and imbalance.



Summary Statistics:

ENGLISH – Class distribution:

```
answer
Business & Management    106
Tax & Accounting          97
Finance                  85
Industry                 78
Technology               54
Government & Controls    28
Name: count, dtype: int64

answer
Business & Management    0.237
Tax & Accounting          0.217
Finance                  0.190
Industry                 0.174
Technology               0.121
Government & Controls    0.062
Name: proportion, dtype: float64
```

GREEK – Class distribution:

```
answer
Business & Management    98
Tax & Accounting          97
Finance                  22
Industry                 9
Government & Controls    8
Technology               1
Name: count, dtype: int64

answer
Business & Management    0.417
Tax & Accounting          0.413
Finance                  0.094
Industry                 0.038
Government & Controls    0.034
Technology               0.004
Name: proportion, dtype: float64
```

SPANISH – Class distribution:

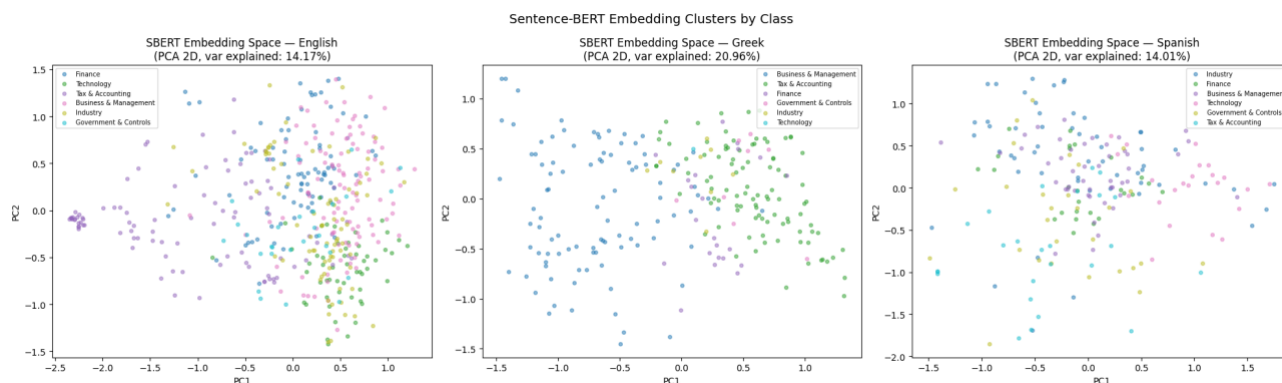
```
answer
Industry                 55
Business & Management    46
Government & Controls    28
Finance                  26
Tax & Accounting          25
Technology               23
Name: count, dtype: int64

answer
Industry                 0.271
Business & Management    0.227
Government & Controls    0.138
Finance                  0.128
Tax & Accounting          0.123
Technology               0.113
Name: proportion, dtype: float64
```

From the Summary statistics, the most important key take way is there is class imbalance and Language imbalance.

- In English , it is fairly balanced across top classes but Government & Controls is under-represented (6.2%).
- In case of Greek, it is heavily skewed towards Business & Management and Tax & Accounting with over 83%.
- In Spanish, this is the most balance out of the three with no class being extremely smaller in size.

SBERT Embedding Clusters by Class (PCA 2D)



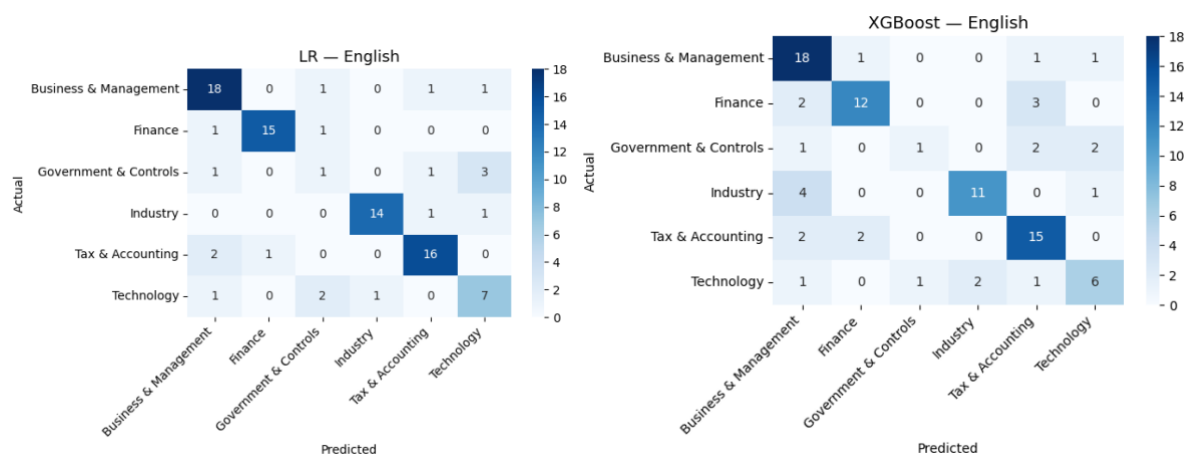
In the above graph, it visualizes sentence BERT embeddings, there are 768 dimensional vectors which reduces into 2D using PCA so we can see a structure where Each dot is a text sample, colors represent domain classes.

If the clusters are more separated, it means SBERT embedding are capturing useful semantic differences between classes. In our results, the Clusters Overlap in many such places which shows why misclassification happens. Till now, it compares to TF IDF SBET embedding which represents a better meaning which is why it under performed when compared to Logistic Regression and XG Boost.

- In English, The points are well spread out with heavy overlap between classes meaning many finance categories share similar wordings.
- In Greek, the plot shows a clear separation compared to English and Spanish as many of them are dominated by a Business & Management, Tax & accounting in Greek data.
- In case of Spanish, it looks more mixed and Scattered with multiple classes overlapping strongly, This shows Spanish texts share common domain terms across classes.

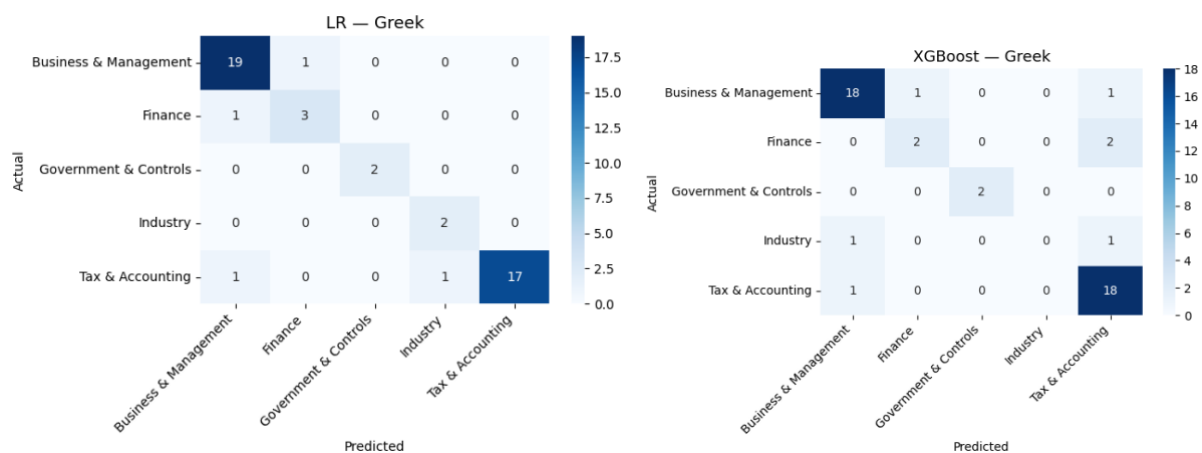
Model Results with SBERT Embeddings:

English (Logistic Regression Vs XG Boost):



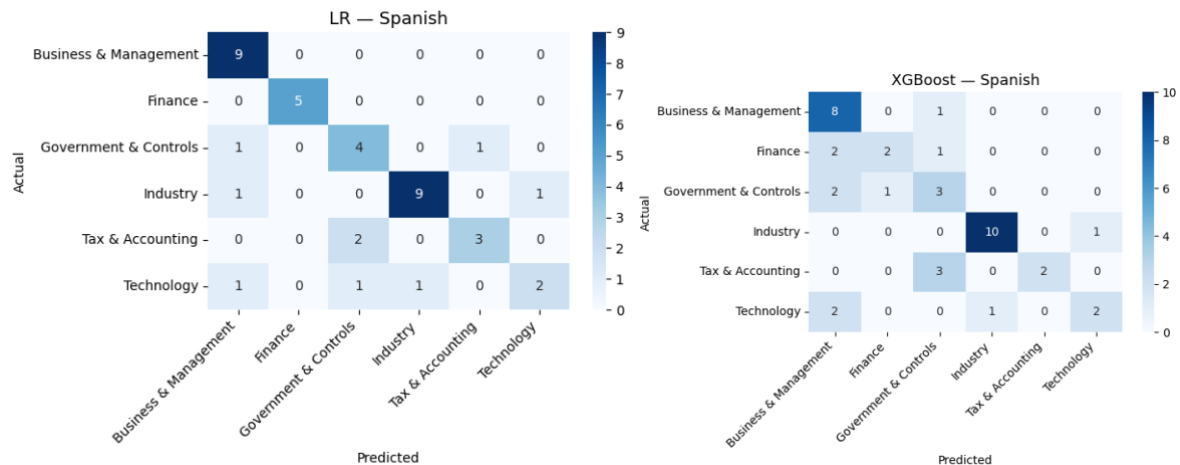
- In English, The logistic regression has a accuracy of 0.78 and F1 macro of 0.70 which performs better than XG Boost Accuracy of 0.75 and F1 macro of 0.67.
- From the above confusion matrix, Logistic Regression is strong on Finance , Industry , Tax and Accounting but it is les on Government & controls with very low recall because it has few samples and overlaps with topics.
- XGBoost shows similar confusion patterns and it also mostly fails to recover from Government & controls which shows both models suffer from minory class imbalance. Overall, Logistic regression is best choice here.

Greek (Logistic Regression Vs XG Boost):



- In Case of Greek, The logistic Regression has values of 0.91 and 0.83 in Accuracy and F1 macro which is higher than XG Boost with accuracy of 9.89 and macro 0.73.
- In the above confusion matrices, Logistic regression predictions are mostly diagonal showing a very strong separation for Business & Management, Tax & Accounting which are biggest Greek Claases.
- In XG boost, it drops in macro F1 because of very small classes which also has very few examples in text.

Spanish (Logistic Regression Vs XG Boost):

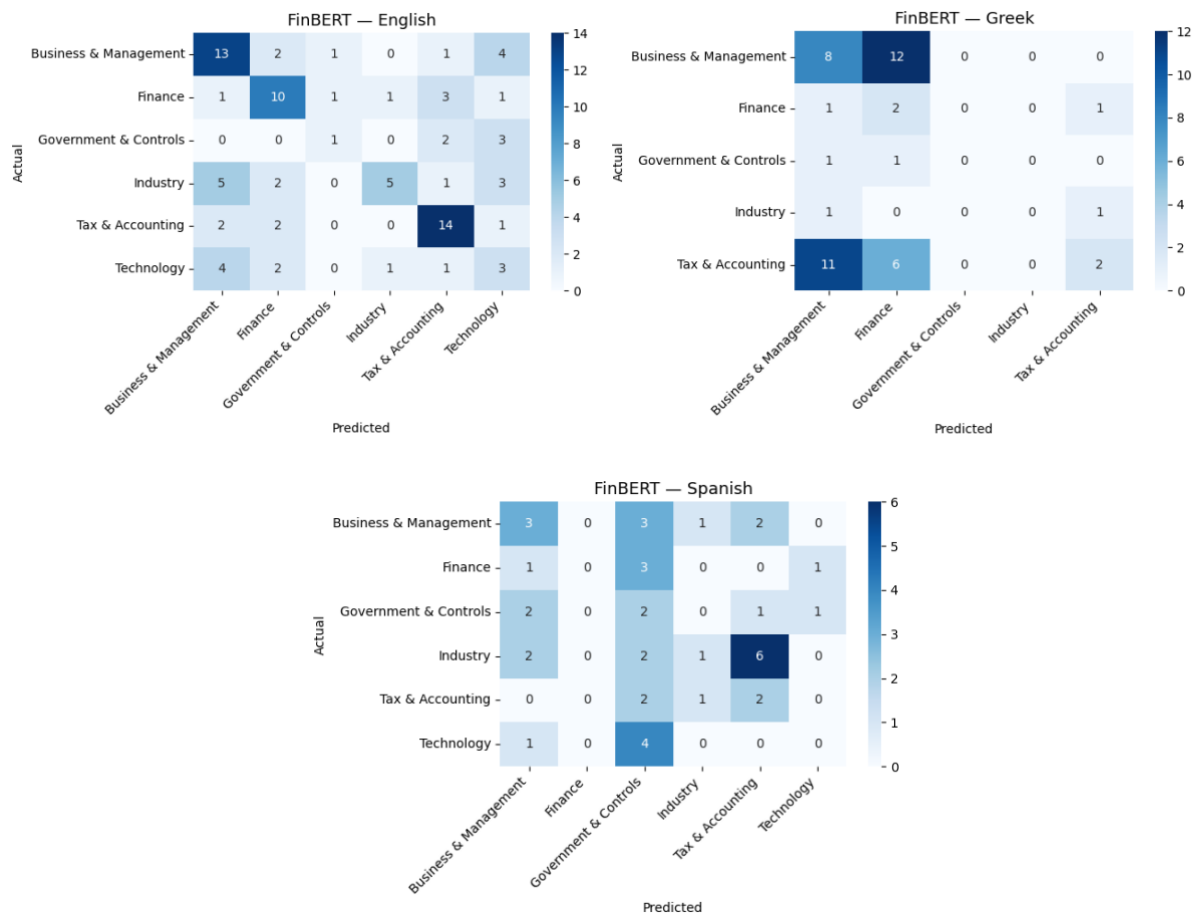


- Logistic Regression with Accuracy of 0.78, F1-macro of 0.74 is much better than XGBoost with Accuracy of 0.6341 and F1-macro of 0.57.
- The Spanish Logistic Regression confusion matrix shows very good predictions for Industry and Finance and decent performance for Business & Management.
- The XG Boost has more off diagonal errors especially among these classes which are semantically closer ones like Government & Controls and Tax & Accounting.

Overall Logistic regression is better across all three languages where LR has consistently higher F1 macro which is more important as dataset is imbalanced. The minority classes like Government & controls has low recall and main weakness in both models.

FinBERT Fine-tuning:

In this FinBERT fine tuning model, we fine tune ProsusAI/finbert separately on English, Greek, and Spanish datasets. FinBERT is a transformer model trained for financial text. In Finbert we split each language dataset into train, test, encodes labels, tokenizes text, loads finBERT model with handling of class imbalance and then train for 5 epochs to evaluate accuracy and macro F1 to plot confusion matrices.



From the above graph for FinBERT english , it is evident that the strongest diagonal cells with correct predictions are Business & Management , Finance and Tax & Accounting. The FinBERT underperforms due to misclassification , technology often goes with Business , Finance where Government & control is weak.

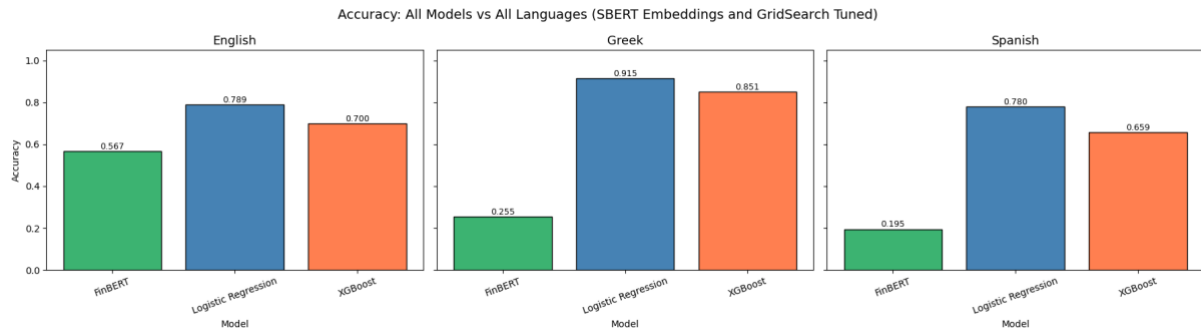
In FinBert Greek, the model predicts mostly Business & Management, Finance where Tax & accounting get predicted as Business Management or Finance , Some classes basically give out not so correct predictions.

In FinBert Spanish, The technology is again predicted as Government & controls, Industry also gets mixed with Tax, Accounting or Government & controls. The model predictions are also scattered and inconsistent.

Although FinBERT is a strong financial domain transformer model, it performed much better on English than Greek or Spanish. The SBERT multilingual embeddings combined with Classical Classifiers give higher performance across all languages.

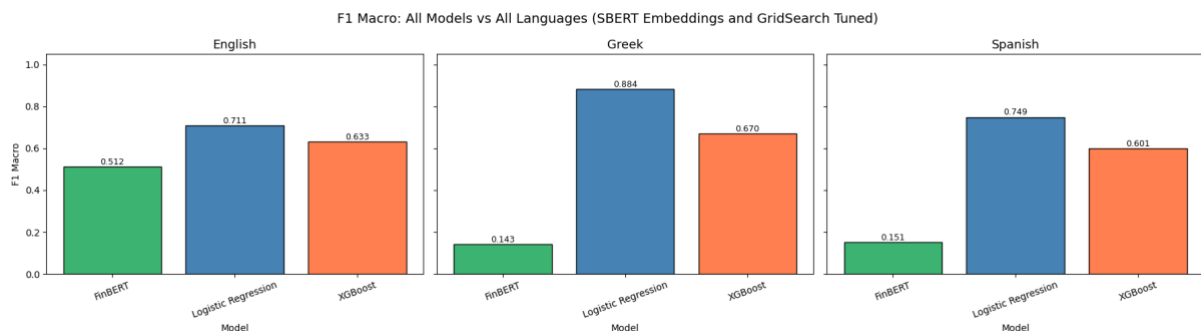
RESULTS:

Accuracy Results and Comparison:



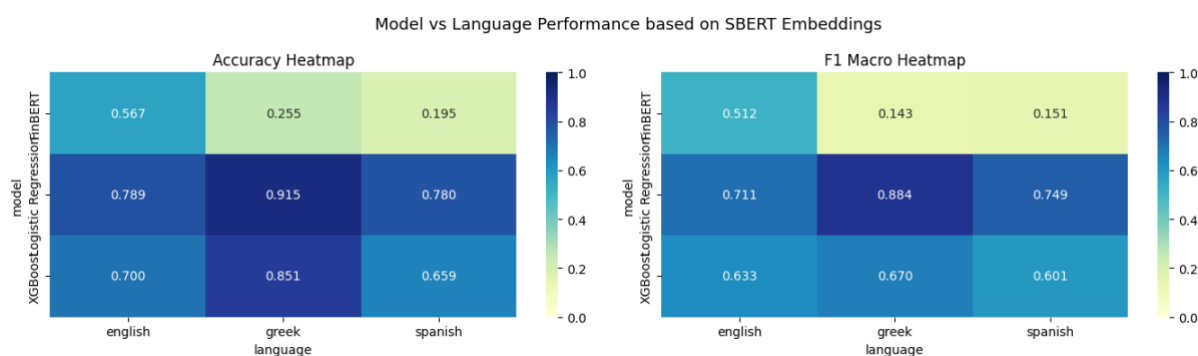
- In English, the Logistic regression is the best with 0.789 and XGBoost is so close with 0.700 with FinBERT as low as 0.567. This indicates the SBERT embeddings worked better than fine tuning FinBERT for English.
- In case of Greek, The logistic regression still performs better with 0.915 Accuracy and XGBosst with 0.851 while FInBERT performs the least. This is same in case of Spanish as well where Spanish has logistic regression of 0.780 and XGBoost of 0.659
- This shows multilingual SBERT embeddings works well with Greek and Spanish but FInBERT struggles as it mainly English focussed.

F1 Macro Results and Comparison:



- From the above comparison graph, The Logistic regression has the best macro F1 of 0.711 with XG Boost of 0.633 and FinBert of 0.512. Here macro F1 works well with each class equally in English.
- For Greek, The Logission Regression is best with 0.884 followed by 0.670 of XGBoost.
- This is same case for Spanish as well with Logistic Regression has higher value of 0.749 , XGBoost with a moderate value of 0.601 and FinBert was very low show how accurate and consisitent with categories is Logistic Regression.
- F1 macros Score is very important as our dataset is very imabalance, High accuracy alone isn't enough, but LR gives the best macro F1 and accuracy so far being more reliable with Embeddings.

Performance Heatmaps with Accuracy and F1 Macro:



From the above heatmaps of Accuracy and F1 macro with all three models, it clearly summarises that Logistic regression row is darkest across language showing higher accuracies and F1 macro scores as well.

The XG Boost is very strong in case of Greek and decent in English but drops in Spanish.

FinBERT has very low scores especially in Spanish and Greek.

The SBERT embeddings are the main reason why performance improved across various languages as SBERT is multilingual and produces comparable semantic vectors where FinBERT did not generalise to Greek and Spanish as its domain is mostly trained with English alone and its tokenizer isn't multilingual like SBERT.

Conclusion:

From this project, We were able to build a multilingual financial Q& A system for Task 1 of FinMMEval which includes domains such as Finance, Industry, Technology, Tax & Accounting, Business & Management, Government & Controls. Initially we started with traditional baseline such as TFIDF , Logistic Regression and then we explored and expanded our scope by adding Greek and Spanish datasets.

After running Exploratory analysis and training, testing and fine tuning models, Sentence Bert multilingual embeddings used with Logistic Regression performed the best overall across various languages. It did get to a strong and consistent results such 0.789 accuracy and 0.70 F1 in English. 0.91 accuracy and 0.83 F1 score in Greek and 0.78 accuracy , 0.74 F1 score in Spanish. This indicate that embeddings work greatly across languages which are effective for short headline style texts. On the other hand, FinBERT tuning performed well only for English showing that finance specific models are mostly training in English.

In Future work, There could be a multilingual transformer classifier which is fine tuned for domain classification instead of FinBERT. The data balance could be further increased using data augmentation , oversampling or adding more samples. SBERT pipeline could be improved by testing different multilingual models and changing larger k values. Also Manually inspect misclassified samples for Government , controls, Finance Industry overlap and add more domain specific keywords during error analysis.

Lessons Learned:

The Class imbalance does matter a lot where minority classes like Government& control are harder to learn and reduce F1 even if accuracy is high. Embedding are more powerful when it comes to multilingual NLP where SBERT multilingual embeddings capture semantic meaning across languages than other translation. Evaluation metric such as Macro F1 was more informative than accuracy as it reflects performance across all categories not just majority classes. More scaling , selectKBest and balanced class weights make LR stable and effective.

Acknowledgements

We would like to thank our Professor Dr. Duoduo liao for guidance and feedback through the project , We also acknowledge the using of HuggingFace Datasets, models, Google colab for computing resources and open source libraries to support our project. Our Team contributed equally where we did split across dataset handling , Exploratory analysis, baseline modelling , multilingual expansion, embedded based modeling and evaluation.

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